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Chemical Kinetics for the Oxidation of Californium(III) Ions with Select Radiation-Induced Inorganic Radicals ($Cl_2^{\bullet-}$ and $SO_4^{\bullet-}$)

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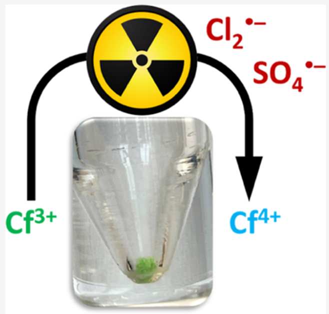
ABSTRACT: Despite the availability of transuranic elements increasing in recent years, our understanding of their most basic and inherent radiation chemistry is limited and yet essential for the accurate interpretation of their physical and chemical properties. Here, we explore the transient interactions between trivalent californium ions (Cf^{3+}) and select inorganic radicals arising from the radiolytic decomposition of common anions and functional group constituents, specifically the dichlorine ($Cl_2^{\bullet-}$) and sulfate ($SO_4^{\bullet-}$) radical anions. Chemical kinetics, as measured using integrated electron pulse radiolysis and transient absorption spectroscopy techniques, are presented for the reactions of these two oxidizing radicals with Cf^{3+} ions. The derived and ionic strength-corrected second-order rate coefficients (k) for these radiation-induced processes are $k(Cf^{3+} + Cl_2^{\bullet-}) = (8.28 \pm 0.61) \times 10^5\ M^{-1}\ s^{-1}$ and $k(Cf^{3+} + SO_4^{\bullet-}) = (9.50 \pm 0.43) \times 10^8\ M^{-1}\ s^{-1}$ under ambient temperature conditions (22 ± 1 °C).



■ INTRODUCTION

The 5*f* elements, known as the actinides, boast many unique physical and chemical properties,[1] including all of their isotopes being inherently unstable with respect to radioactive decay. Despite this, our understanding of their radiation-induced chemistry in the condensed phase is limited.[2−4] Studies in this area have predominantly focused on the radiation-induced behavior of the early (thorium and uranium) to mid (neptunium to curium) actinides, owing to their availability and role in nuclear energy and weapon technologies.[4] As such, much of our actinide radiation chemistry knowledge is constrained to those applied conditions. For example, the behavior of actinide ions in irradiated aqueous nitric acid solutions has been well studied due to this solvent's role in separation technologies.[4−9] However, this applied focus has created a large radiation chemistry knowledge gap, as fundamental actinide studies utilize many different solutes and solvents as compared to the nuclear industry, to elucidate the nature and role of the 5*f* electrons in chemical bonding. As such, the radiation-induced chemistry of these fundamental systems and their interactions with the actinides must be explored to provide a comprehensive understanding of actinide chemistry, and to ensure accurate interpretation of their physical and chemical properties, as radiolytic processes can lead to unanticipated phenomena. This endeavor is especially important for the less-studied late actinides (berkelium onward), for which only a handful of radiation studies exist,[10−12] despite them becoming more readily available to research communities.

Actinide chlorides are one of the most common starting materials used in fundamental actinide studies,[13−20] and as such, the chloride anion (Cl^-) is often present in either the inner or outer coordination sphere of many reported actinide crystal structures,[21−26] including californium.[25] Furthermore, hydrochloric acid (HCl) and chlorinated hydrocarbons such as dichloromethane (DCM) are widely used for the purification and recycling of transuranic elements and as solvents in subsequent studies.[27−29] Consequently, there are many scenarios in which Cl^- ions and chlorinated compounds may be exposed to ionizing radiation fields arising from the radioactive decay of the actinides and their radioactive daughter nuclides. The radiation emitted induces ionization and excitation of the surrounding medium, with the subsequent formation of reactive radiolysis products. These primary radiolysis products have the capacity to interact with any remaining actinide ions to drive changes in their redox distribution and the potential formation of transient non-equilibrium actinide oxidation states. For example, trivalent californium ions (Cf^{3+}) can be transiently reduced and oxidized to the corresponding di- (Cf^{2+}) and tetravalent states (Cf^{4+}) by reaction with the hydrated electron (e_{aq}^-) and hydroxyl radical ($^{\bullet}OH$) arising from the direct radiolysis (⇝) of water[12]

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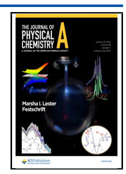

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$$H_2O \rightsquigarrow H_2O*, e_{aq}^-, H^\bullet, {}^\bullet OH, H_2O_2, H_2, H_{aq}^+ \quad (1)$$

$$Cf^{3+} + e_{aq}^- \rightarrow Cf^{2+}$$
$$k = (3.63 \pm 0.14) \times 10^{10}\ M^{-1}\ s^{-1} \quad (2)$$

$$Cf^{3+} + {}^\bullet OH \rightarrow Cf^{4+} + OH^-$$
$$k = (2.07 \pm 0.12) \times 10^{8}\ M^{-1}\ s^{-1}. \quad (3)$$

Both of these nonequilibrium californium oxidation states were found to exhibit lifetimes on the order of tens of microseconds, which is more than sufficient time to promote currently unexplored actinide redox chemistry.

With regard to Cl^- ions, their direct and indirect radiolysis can promote the formation of chlorine radicals ($Cl^\bullet$) and the subsequent generation of the dichlorine radical anion ($Cl_2^{\bullet-}$)[30−34]

$$Cl^- \rightsquigarrow e_{aq}^- + Cl^\bullet \quad (4)$$

$$Cl^- + {}^\bullet OH \rightleftharpoons ClOH^{\bullet-} \quad (5)$$

$$ClOH^{\bullet-} + H_{aq}^+ \rightleftharpoons Cl^\bullet + H_2O, \quad (6)$$

$$Cl^\bullet + Cl^- \rightleftharpoons Cl_2^{\bullet-} \quad (7)$$

the fate of which is mediated by its disproportionation (eq 8)[35−50] and electron transfer reactions with other solutes (eq 9)[2−4]

$$Cl_2^{\bullet-} + Cl_2^{\bullet-} \rightarrow Cl_3^- + Cl^-$$
$$k = 0.7 - 9.0 \times 10^{9}\ M^{-1}\ s^{-1} \quad (8)$$

$$\text{solute} + Cl_2^{\bullet-} \rightarrow \text{solute}^{\bullet+} + 2Cl^- \quad (9)$$

A similar pathway is proposed for the radiolysis of chlorinated organic media, exemplified here with DCM (CH_2Cl_2), where solvated electrons (e_s^-) excite DCM ($CH_2Cl_2^{*-}$) and then liberate Cl^- ions (eqs 10 and 11)[27]

$$CH_2Cl_2 + e_s^- \rightarrow CH_2Cl_2^{*-} \quad (10)$$

$$CH_2Cl_2^{*-} \rightarrow CH_2Cl^\bullet + Cl^- \quad (11)$$

which may then go on to react via eqs 4−7 to yield additional $Cl_2^{\bullet-}$ radical anions. Irrespective of the source, the $Cl_2^{\bullet-}$ radical anion is a strongly oxidizing species ($E° = +2.13$ V),[51] which has been shown to react with a number of actinide ions[4] and thus has the capacity to influence actinide studies that employ chloride-containing chemicals. For example, the current renaissance in molten salt technologies has catalyzed extensive applied and fundamental studies into the behavior of actinide ions in high-temperature halide salt mixtures, including chlorides.[52]

In addition to chloride systems, some of the earliest fundamental actinide crystallography studies were performed with simple sulfate salts (SO_4^{2-}).[53−57] These experiments were some of the first steps toward providing a deep understanding of the unique chemistry exhibited by transuranic elements. More recently, crystallographic studies have demonstrated the formation of direct actinide−sulfate coordination structures in the form of 3D actinide sulfate networks[58] and bridging neptunyl centers.[59] Overall, actinide sulfates have provided invaluable information on the unusual bonding modes and coordination environments available to the transuranic elements.[60,61]

With that being said, the influence of radiolysis on these actinide sulfate systems has not been well studied. Irradiation of sulfur-containing ions and molecules can lead to the formation of various sulfur radicals.[62] For example, the reaction of the ${}^\bullet OH$ radical with hydrogen sulfate (HSO_4^-) can yield the oxidizing sulfate radical anion ($SO_4^{\bullet-}$, $E° = +2.5−3.1$ V)[63−66]

$$HSO_4^- + {}^\bullet OH \rightarrow SO_4^{\bullet-} + H_2O$$
$$k = 0.7 - 1.7 \times 10^{6}\ M^{-1}\ s^{-1} \quad (12)$$

As with the $Cl_2^{\bullet-}$ radical anion, the decay of the $SO_4^{\bullet-}$ radical anion is characterized by its rate of disproportionation and reaction with other solutes, including several actinide ions[2−4,65−74]

$$SO_4^{\bullet-} + SO_4^{\bullet-} \rightarrow S_2O_8^{2-}$$
$$k = 0.2 - 1.8 \times 10^{9}\ M^{-1}\ s^{-1} \quad (13)$$

$$\text{solute} + SO_4^{\bullet-} \rightarrow \text{solute}^{\bullet+} + SO_4^{2-} \quad (14)$$

These solute reactions can take the form of electron transfer, hydrogen atom abstraction, and addition to unsaturated chemical bonds, again providing the opportunity for currently unanticipated, nonequilibrium actinide chemistry.

With the capacity of these two radical anions to promote redox reactions with the early-to-mid actinides,[4] combined with the prevalence of their precursors in actinide studies, it is important to understand their interactions with the late actinides in support of a comprehensive understanding of radiation-induced actinide redox chemistry. As such, we report the kinetics for the reaction of these two oxidizing ($Cl_2^{\bullet-}$ and $SO_4^{\bullet-}$) radicals with Cf^{3+} ions in aqueous solution, as measured using integrated electron pulse radiolysis and transient absorption spectroscopy techniques.

■ METHODOLOGY

Caution! The californium-249 solutions employed in this work were highly radioactive. Handling was performed in dedicated radiological and nuclear facilities using well-established radiological safety protocols.

Chemicals. Californium-249 (^{249}Cf, $\tau_{1/2}$ = 351 years, E_α = 6.30 MeV) was initially sourced as its chloride salt ($CfCl_3$) from the Isotope Development and Production for Research and Applications Program through the Radiochemical Engineering and Development Center at Oak Ridge National Laboratory. Ammonium hydroxide (NH_4OH, ≥99.999% trace metal basis, 28% NH_3 in water), deionized water (DI H_2O, ultrapure HPLC grade), ethanol (≥99.5%), formic acid (≥98%), hydrochloric acid (HCl, 37 wt %), hydrogen peroxide (H_2O_2, 30 wt %), oxalic acid (98%), perchloric acid ($HClO_4$, ≥99.999% trace metal basis), potassium persulfate ($K_2S_2O_8$, 99.99% trace metal basis), sodium chloride (NaCl, 99.999% trace metal basis), and sodium hydroxide (NaOH, 99.99% trace metal basis) were purchased from MilliporeSigma (Burlington, Massachusetts). Compressed argon (Ar) and nitrous oxide (N_2O) with purities of ≥99.5% were purchased from Airgas (Radnor, Pennsylvania). Unless otherwise stated, all chemicals were used without further purification.

Californium Purification. The californium used in this study was recovered from previous experiments. To avoid kinetic interference from accumulated decay products and potential contaminants, the californium was first repurified at

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Florida State University. This involved the dissolution of californium-containing solids in 6 M HCl. The resulting acidic solution was then passed through an Eichrom (Eichrom, Lisle, Illinois) prefilter resin (100−150 μm mesh) to remove any residual organic material. The resin was preconditioned with three bed volumes of DI H_2O and three bed volumes of 6 M HCl. Once >99% of the Cf^{3+} was eluted with 6 M HCl, as determined by dosimetry using a 451P radiation detector (Fluke Biomedical, Everett, Washington), the resulting Cf^{3+} eluent was dried to a residue using a 250 W heat lamp and a slow stream of nitrogen gas. To remove cationic impurities, the Cf^{3+} residue was dissolved in 0.1 M HCl and then loaded onto a cation exchange resin (AG50W-X8, 100−150 μm mesh, Bio-Rad Laboratories, Hercules, California) that had been preconditioned sequentially with three bed volumes of DI H_2O, three bed volumes of 6 M HCl, and three bed volumes of 0.1 M HCl. Once fixed on the column, the Cf^{3+} was rinsed with excess 0.1 M HCl and then 2 M HCl, before eluting the Cf^{3+} with excess 6 M HCl. As mentioned before, once >99% of the Cf^{3+} had been recovered, the resulting eluent was dried to a residue. To remove anionic impurities, an anion exchange resin (AG MP-1M, 50−100 μm mesh, Bio-Rad Laboratories, Hercules, California) was employed and preconditioned with three bed volumes of DI H_2O and three bed volumes of 6 M HCl. The californium residue was again dissolved, this time in minimal 6 M HCl, and then loaded onto the preconditioned anion column. The column was then washed with 6 M HCl until >99% of the Cf^{3+} was eluted, leaving any remaining potential contaminants bound to the column. This final eluent was again dried to a residue, redissolved in 2 M HCl, and subsequently recrystallized as the corresponding oxalate by the excess addition of a saturated oxalic acid/2 M HCl solution. The californium oxalate was then decomposed by dissolution in HNO_3 at 120 °C in a KSL-1100X-S-UL box furnace (MTI Corporation, Richmond, California) over 12 h. The resulting residue was then redissolved in 2 M HCl for quantification by absorption spectroscopy using a Cary 6000*i* UV−vis−NIR instrument (Agilent, Santa Clara, California). A final recovery yield of 5.99 mg californium was calculated from averages derived from the element's 442, 602, and 673 nm absorption bands[17,75] for 1 mL of solution with a 0.4 cm optical path length in a dual-path length, GL15 Spectrosil Far UV quartz cell (Starna Scientific Ltd., Ilford, UK). The resulting spectrum is shown in SI Figure S1 alongside that previously reported by Conway et al. for comparison.[75]

From this 5.99 mg of Cf^{3+}/2 M HCl stock solution, 2 mg worth of Cf^{3+} content (333.4 μL) was isolated, dried to residue, and redissolved in 3 mM $HClO_4$ solution. This Cf/$HClO_4$ stock solution was then used in the preparation of all sample solutions to be irradiated. Samples were prepared by the serial dilution of 410 μL of Cf/$HClO_4$ stock solution into one of two aqueous solutions designed for the isolation of the $Cl_2^{\bullet-}$ and $SO_4^{\bullet-}$ radical anions and the direct measurement of their associated reaction kinetics with the Cf^{3+} ion:[78]

- Dichlorine radical anion: 1 M NaCl dissolved in aerated, aqueous 0.1 M $K_2S_2O_8$/3 mM $HClO_4$ solution. Direct decay kinetics of the $Cl_2^{\bullet-}$ radical anion monitored at 350 nm.
- Sulfate radical anion: 0.1 M $K_2S_2O_8$ dissolved in aerated, aqueous 3.0 mM $HClO_4$ solution. Direct decay kinetics of the $SO_4^{\bullet-}$ radical anion monitored at 450 nm.

The final Cf^{3+} ion sample concentrations were 0, 0.2, 0.4, 0.6, 0.8, and 1.0 mM in each radical stock solution. These samples were sealed in 1.0 cm optical path length, screwcap-sealed, spectrosil quartz, semimicro, Starna Scientific Ltd. (Ilford, United Kingdom) cuvettes for electron pulse irradiation.

The isolation of these two radicals was based on the reaction of the e_{aq}^- reaction with $S_2O_8^{2-}$ in aqueous solution[64]

$$e_{aq}^- + S_2O_8^{2-} \rightarrow SO_4^{2-} + SO_4^{\bullet-} \quad k = 1.2 \times 10^{10}\ M^{-1}\ s^{-1} \tag{15}$$

which quantitatively produces the $SO_4^{\bullet-}$ radical anion that then reacts with added Cl^- ions[77]

$$SO_4^{\bullet-} + Cl^- \rightarrow Cl^\bullet + SO_4^{2-} \quad k = 2.8 \times 10^{8}\ M^{-1}\ s^{-1} \tag{16}$$

to give $Cl^\bullet$ radicals that ultimately result in the $Cl_2^{\bullet-}$ radical anion (eq 7). Formation of the potentially competing hypochlorite radical anion ($ClOH^{\bullet-}$) was minimized through the scavenging of the initially formed $^\bullet OH$ radicals by the $S_2O_8^{2-}$ ions[78]

$$^\bullet OH + S_2O_8^{2-} \rightarrow OH^- + S_2O_8^{\bullet-} \quad k = 1.2 \times 10^{7}\ M^{-1}\ s^{-1} \tag{17}$$

Time-Resolved Picosecond Electron Pulse Irradiations. Kinetic measurements for the reaction of Cf^{3+} ion with $Cl_2^{\bullet-}$ and $SO_4^{\bullet-}$ radical anions under ambient temperature conditions were performed by integrated electron pulse radiolysis and transient absorption detection using the Brookhaven National Laboratory (BNL) Laser Electron Accelerator Facility (LEAF) system.[79] As discussed above, sample solutions were designed to facilitate the direct production of a given radical and its measured decay at a specific wavelength accessible to the LEAF system, 300−1000 nm, as selected by 10 nm bandpass interference filters (hard coated, Edmund Optics). Time-resolved changes in radical absorption decay as a function of Cf^{3+} ion concentration were followed using a pulsed xenon arc lamp, Photon Technology International (London, Ontario, California) lamp housing, Ushio (Tokyo, Japan) bulb, a custom-made pulser circuit, and a FND-100 silicon diode detector. The resulting signals were subsequently digitized using a Teledyne LeCroy (Chestnut Ridge, New York City) WaveRunner 640Zi 4 GHz 8-bit oscilloscope. Chemical dosimetry was performed using argon-sparged standard solution, 1.0 M NaOH, and 20 vol % ethanol in Millipore DI water, by measuring the decay of the e_s^- at 650 nm.[80] This approach afforded an average dose per pulse of ~9.8 Gy.

As both $Cl_2^{\bullet-}$ and $SO_4^{\bullet-}$ radicals are negatively charged and the target Cf^{3+} ion is positively charged, the derived second-order rate coefficients (k) were corrected for the ionic strength of each sample solution to account for the effects of coulombic attraction[81]

$$\log(k_0) = \log(k) - 2Z_\alpha Z_\beta[A\mu^{1/2}/(1 + \mu^{1/2})] \tag{18}$$

where k_0 is the rate coefficient ($M^{-1}\ s^{-1}$) at zero ionic strength, k is the rate coefficient measured here at a total ionic strength μ (mol L^{-1}), Z_α and Z_β are the respective charges of the radical

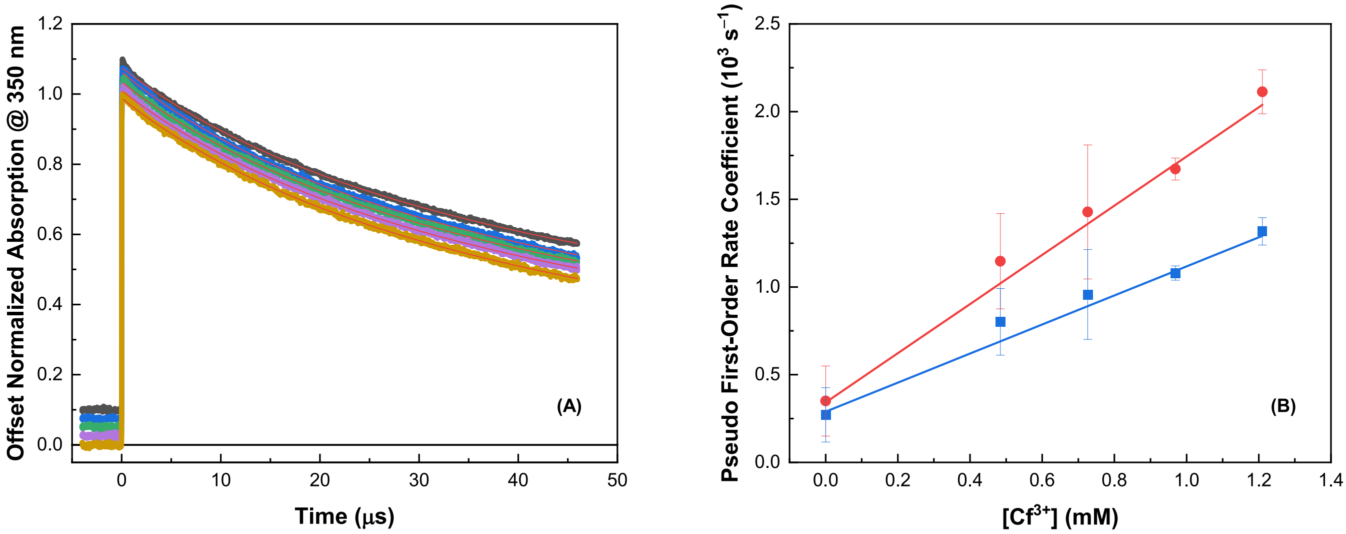

Figure 1. (**A**) Offset dose- and absorption-normalized $Cl_2^{\bullet-}$ radical anion decays at 350 nm for zero (**black**), 0.48 (**blue**), 0.73 (**green**), 0.97 (**purple**), and 1.21 (**gold**) mM Cf^{3+} in electron pulse-irradiated aqueous solutions containing 1 M NaCl and 0.1 M $K_2S_2O_8$ in 3 mM $HClO_4$ at 22 ± 1 °C. Solid red lines are mixed-order fits to data. (**B**) Second-order rate coefficient determination by plotting derived first-order component values against Cf^{3+}concentration with (**blue** ■) and without (**red** ●) ionic strength corrections. This afforded an ionic strength-corrected value of $k(Cf^{3+} + Cl_2^{\bullet-}) = (8.28 \pm 0.61) \times 10^5\ M^{-1}\ s^{-1}$, $R^2 = 0.98$.

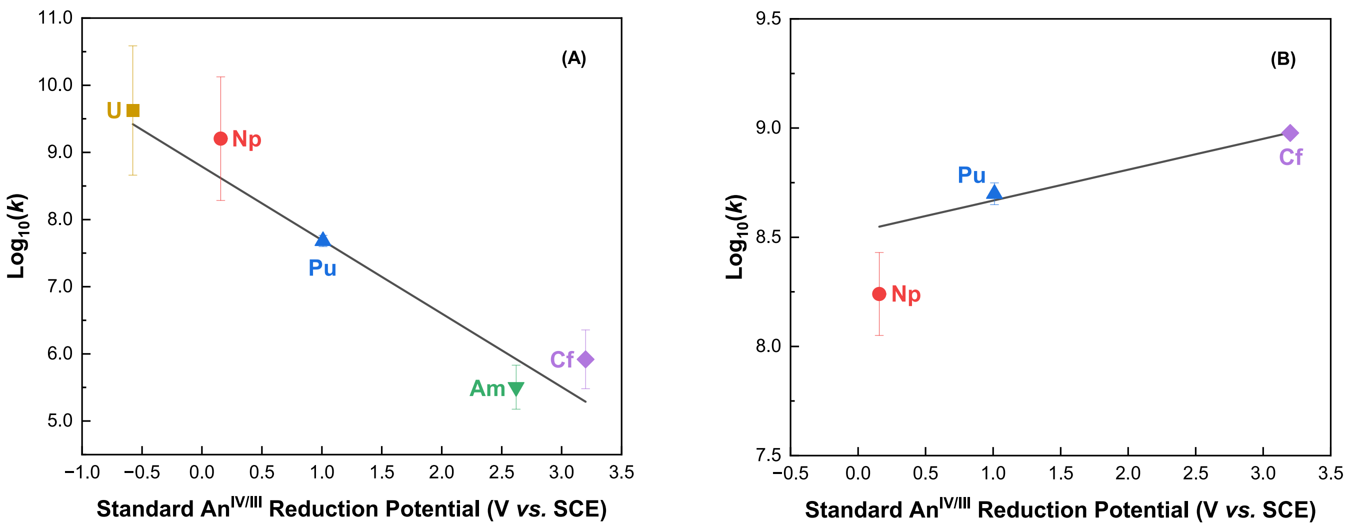

Figure 2. Reaction rates for the $Cl_2^{\bullet-}$ (**A**) and $SO_4^{\bullet-}$ (**B**) radical anions with trivalent actinide ions from this work (Cf^{3+}) and previously published values[4] vs the standard reduction potentials for the corresponding IV/III couples.[76]

anions ($Z_\alpha = -1$) and Cf^{3+} ($Z_\beta = +3$), respectively, and A is the temperature-dependent Debye−Hückel constant[82]

$$A = \frac{e^3(2000 \times N_A)^{1/2}}{2.303(8\pi)(\varepsilon_0 \varepsilon k_B T)^{3/2}} \tag{19}$$

where e is the charge of an electron (1.602×10^{-19} C), N_A is Avogadro's number (6.022×10^{23} mol^{-1}), ε_0 is the vacuum permittivity (8.854×10^{-12} F m^{-1}), ε is the dielectric constant for water (78.84), k_B is the Boltzmann constant (1.380×10^{-23} J K^{-1}), and T is the temperature (K).

Quoted rate coefficient errors are the standard deviation of each decay trace, which are the average of three separate decay traces composed of eight measured pulses per sample.

■ RESULTS AND DISCUSSION

Chemical kinetics for the reaction of the Cf^{3+} ion with the $Cl_2^{\bullet-}$ radical anion are shown in Figure 1. It is evident from the small changes in the decay rate of the $Cl_2^{\bullet-}$ radical anion with increasing Cf^{3+} concentration (Figure 1A) that this reaction is slow and energetically unfavorable. This was to be expected given the ~1 V difference in their respective oxidation potentials in acidic aqueous solution: $E°(Cf^{3+}) = +3.2$ V vs $E°(Cl_2^{\bullet-}) = +2.13$ V.[51,76] Despite this energetic difference, pseudo-first-order rate coefficients (k') were derived from the mixed pseudo-first and second-order fits to the kinetic decay data shown in Figure 1A based on the equation[83]

$$\frac{\mathrm{Abs}*k_X}{k_X*e^{(k_X t)} - 2*\mathrm{Abs}*k_Y*(1 - e^{(k_X t)})} + \mathrm{B}, \tag{20}$$

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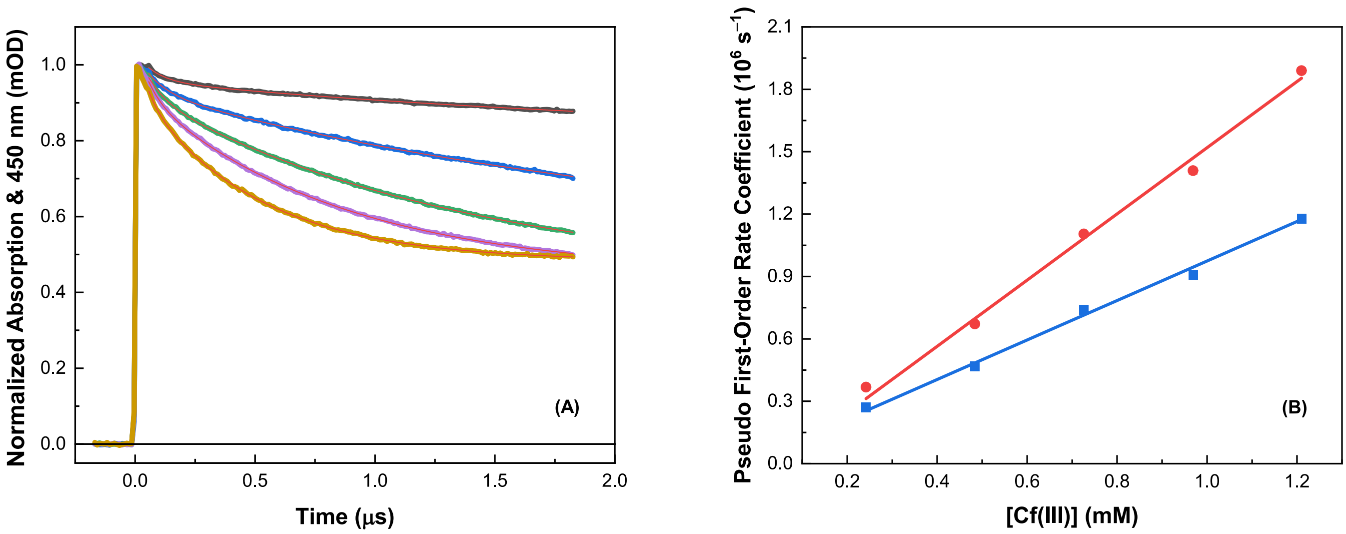

Figure 3. (**A**) Dose- and absorption-normalized $SO_4^{\bullet -}$ radical anion decays at 450 nm for zero (**black**), 0.24 (**blue**), 0.48 (**green**), 0.73 (**purple**), and 1.21 (**gold**) mM Cf(III) in electron pulse-irradiated aqueous solutions containing 0.1 M $K_2S_2O_8$ in 3.0 mM $HClO_4$ at 22 ± 1 °C. Solid red lines are mixed-order fits to data. (**B**) Second-order rate coefficient determination plotting derived first-order component values against solute concentration with (**blue ■**) and without (**red ●**) ionic strength corrections. This afforded an ionic strength-corrected value of $k(Cf^{3+} + SO_4^{\bullet -})$ = (9.50 ± 0.43) × 10^8 M^{-1} s^{-1}, R^2 = 0.994. Note that error bars are smaller than the presented data points.

where Abs is the initial $Cl_2^{\bullet -}$ radical anion absorption, B is the limiting product absorption (baseline shift), k_X is the pseudo-first-order component (s^{-1}) corresponding to the reaction of the Cf^{3+} ion with the $Cl_2^{\bullet -}$ radical anion (eq 9), and k_Y is the second-order component (M^{-1} s^{-1}) for the corresponding disproportionation reaction shown in eq 8. These pseudo-first-order values were then used to calculate an overall second-order rate coefficient of k = (8.28 ± 0.61) × 10^5 M^{-1} s^{-1} (R^2 = 0.98) for the reaction of the Cf^{3+} ion with the $Cl_2^{\bullet -}$ radical anion

$$Cf^{3+} + Cl_2^{\bullet -} \rightarrow Cf^{4+} + 2Cl^- \tag{21}$$

as shown in Figure 1B. This value is significantly slower than those previously measured for the reaction of the Cf^{3+} ion with the $^{\bullet}OH$ and $NO_3^{\bullet}$ radicals: k = (2.07 ± 0.43) × 10^8 and (1.78 ± 0.19) × 10^8 M^{-1} s^{-1}, respectively.[12] Consequently, oxidation by the $Cl_2^{\bullet -}$ radical anion is not expected to play a significant role in the radiation chemistry of californium chloride studies.

Further support for this finding can be found by comparing this newly measured rate coefficient with those previously reported for the reaction of the $Cl_2^{\bullet -}$ radical anion with other trivalent actinides, as shown in Figure 2A. Our rate value for Cf^{3+} oxidation agrees well with the expected free energy relationship prediction, i.e., single electron transfer to the $Cl_2^{\bullet -}$ radical anion becomes slower and less energetically favorable with increasing reduction potential.

Chemical kinetics for the reaction of the Cf^{3+} ion with the $SO_4^{\bullet -}$ radical anion are shown in Figure 3. Note that for the highest concentration of Cf^{3+} used in this study, the decay of the $SO_4^{\bullet -}$ radical anion's absorption clearly does not return to the original baseline (Figure 3A). This is indicative of the presence of another, longer-lived, radiolysis product, the ingrowth of which occurs within the time frame of our experiments. This additional species was not evident in the absence of Cf^{3+} ions, nor can it be attributed to the Cf^{4+} oxidation product, as this species has only minimal absorption at 450 nm.[12] A potential candidate for this species is the oxygen adduct of the persulfate peroxyl radical anion, $(S_2O_8 \bullet O_2)^{\bullet -}$

$$S_2O_8^{\bullet -} + O_2 \rightarrow (S_2O_8 \bullet O_2)^{\bullet -} \tag{22}$$

formed as a subsequent product of the oxidation of $S_2O_8^{2-}$ by the $^{\bullet}OH$ radical (eq 17).[84] Although, no rate coefficient has been reported for eq 22, kinetic measurements at 450 nm in air-saturated solution have demonstrated that adduct formation is complete within ~500 ns.[84] This growth corresponds to a fast formation rate coefficient of $k \sim 10^9$–10^{10} M^{-1} s^{-1}. This magnitude is consistent with the kinetics typically observed for carbon-centered radical peroxyl formation, which exhibit rate coefficients of around 3 × 10^9 M^{-1} s^{-1}.[64] The transient spectrum reported for the $(S_2O_8 \bullet O_2)^{\bullet -}$ adduct shows a peak at 430 nm, with a relative intensity of about 3–4× that of the $SO_4^{\bullet -}$ radical anion and a half-life of ~1 ms.[84]

Regardless of the identity of the additional species, its contribution to the decay of the $SO_4^{\bullet -}$ radical anion was accounted for by the fitted k_Y value in eq 20. After fitting the data in Figure 3A for 0 mM Cf^{3+} using eq 20, the resulting k_Y value ($2k_y$ = (2.70 ± 0.18) × 10^8 M^{-1} s^{-1}) was then held constant when fitting the decays for 0.24–1.21 mM Cf^{3+}. As such, the growth kinetics and limiting baseline absorption afforded by the additional species had a negligible effect upon our derived second-order rate coefficient (Figure 3B) for the reaction of Cf^{3+} with the $SO_4^{\bullet -}$ radical anion, k = (9.50 ± 0.43) × 10^8 M^{-1} s^{-1}. The resulting value is significantly faster than those previously measured for the only other three oxidizing radical reactions with the Cf^{3+} ion, i.e., $^{\bullet}OH$, $NO_3^{\bullet}$, and $Cl_2^{\bullet -}$, and is to be expected given their respective reduction potentials: $E°(SO_4^{\bullet -})$ = +2.5–3.1 V $\gtrsim$ $E°(^{\bullet}OH)$ = +2.7 V > $E°(^{\bullet}NO_3)$ = +2.3–2.6 V > $E°(Cl_2^{\bullet -})$ = +2.13 V.[51,63,64,85]

Our new $SO_4^{\bullet -}$ radical anion rate coefficient is also consistent with the trend exhibited by the limited values available for other trivalent actinides, as summarized in Figure 2B as a function of their respective redox potentials.

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Interestingly, the free energy relationship for the $SO_4^{\bullet-}$ radical anion reaction shows a slight increase in the rate with increasing redox potential (Figure 2B). By comparison with the $Cl_2^{\bullet-}$ radical anion (Figure 2A) and the available trivalent actinide ion values for reaction with $^{\bullet}OH$ and $NO_3^{\bullet}$ radicals (SI, Figure S2),[4,12,86] this difference in free energy behavior implies that actinide oxidation by the $SO_4^{\bullet-}$ radical anion does not proceed via a simple one electron transfer mechanism. An alternative reaction mechanism could be via initial adduct formation, which would exploit the increasing covalency of the actinide ions as the series is traversed. With that being said, we are presently pursuing follow-up electronic structure calculations to resolve this mechanistic discrepancy, in addition to determining the missing values for trivalent uranium, americium, and curium.

Due to the slowness of the $Cl_2^{\bullet-}$ radical anion decay (Figure 1A) and both the $Cl_2^{\bullet-}$ and $SO_4^{\bullet-}$ radical anions' inherent broad absorption bands in the UV region,[87] we were unable to record any spectral information for the Cf^{4+} ion product in these solutions using the current detection system.[12]

■ CONCLUSIONS

In summary, we report new, ambient temperature (22 ± 1 °C), second-order rate coefficients for the reaction of the Cf^{3+} ion with $Cl_2^{\bullet-}$ and $SO_4^{\bullet-}$ radical anions, as measured by integrated electron pulse radiolysis and transient absorption spectroscopy. Our measurements indicate that the $Cl_2^{\bullet-}$ radical anion reaction is slow (k = (8.28 ± 0.61) × 10^5 M^{-1} s^{-1}) and energetically unfavorable and thus unlikely to impact the outcome and interpretation of fundamental californium studies in the presence of Cl^- ions and chlorinated compounds. On the other hand, Cf^{3+} ion oxidation by reaction with the $SO_4^{\bullet-}$ radical anion is significantly faster (k = (9.50 ± 0.43) × 10^8 M^{-1} s^{-1}) than with other common oxidizing radicals ($^{\bullet}OH$ and $NO_3^{\bullet}$) and surprisingly faster than for other trivalent actinide ions. Consequently, $SO_4^{\bullet-}$ radical anion-driven oxidation is expected to play a significant role in californium studies that utilize sulfur compounds. Additionally, this finding opens avenues for the potential use of the $SO_4^{\bullet-}$ radical anion as a high-efficiency oxidizing agent in the exploration of non-equilibrium actinide oxidation states, such as the Cf^{4+} ion.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpca.3c07404.

Absorption spectra of Cf^{3+} in 2 M HCl (blue, this work) and in 2 M $DClO_4$ (red, modified from reference 1) (Figure S1); second-order rate coefficients for the reaction of $^{\bullet}OH$ (**A**) and $NO_3^{\bullet}$ (**B**) radicals with trivalent actinide ions from this work (Cf^{3+}) and previously published values[2−4] versus the standard reduction potentials for the corresponding IV/III couples (Figure S2) (PDF)

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All the authors have accepted responsibility for the entire content of this submitted article and approved submission.

Notes

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